MTH2010 Algebra

Graded Formative Sheet 1

For the tutorials in Week 4 prepare questions 2 and 7.
Please submit your solution to Question 4 via ELE by noon on Wednesday 7 October 2026.

1. **(a)** **(i)** Find the subgroup of $(\mathbf{Z}_8, +)$ with cardinality 4 and list the left cosets of this subgroup in $\mathbf{Z}_8$.

(ii) Find the subgroup of $(\mathbf{Z}_{12}, +)$ with cardinality 6 and list the left cosets of this subgroup in $\mathbf{Z}_{12}$.

(b) Let

$$D_8 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$$

be the dihedral group with 8 elements, with $r^4 = s^2 = 1$ and $sr = r^3 s$.
List the left cosets of the subgroups $\langle sr^2 \rangle$, and $\langle r \rangle$, generated by sr^2, and r, respectively, in D_8.

2. Let k and n be integers ≥ 2 with k dividing n. Let

$$D_{2n} = \{1, r, \ldots, r^{n-1}, s, sr, \ldots, sr^{n-1}\}$$

be the dihedral group with $2n$ elements and $H = \langle r^k \rangle$ the subgroup of D_{2n} generated by r^k.

(a) Construct a surjective homomorphism from D_{2n} to the dihedral group D_{2k} with $2k$ elements with kernel H.

(b) Deduce that the quotient group D_{2n}/H is isomorphic to D_{2k}.

3. Show that there is no surjective homomorphism from A_4 to $\mathbb{Z}/2\mathbb{Z}$.
Is there a surjective homomorphism from A_4 to $\mathbb{Z}/3\mathbb{Z}$? (justify your answer and construct such a homomorphism if it exists).

4. **(a)** Let G be a group and let H and K be subgroups of G. Show that the intersection $H \cap K$ of H and K is a subgroup of G. Prove that if H and K are normal subgroups then $H \cap K$ is also a normal subgroup of G.

(b) If N is a normal subgroup of a group G and H any subgroup of G show that $N \cap H$ is a normal subgroup of H.

5. **(a)** Let A and B be groups. Prove that the subset

$$\tilde{A} = \{(a, e_B) : a \in A\}$$

of $A \times B$ is a subgroup of $A \times B$. Prove that $\tilde{A}$ is a normal subgroup. Show that the quotient group $(A \times B)/\tilde{A}$ is isomorphic to B.

(b) Let A be an abelian group and let B be a subgroup of A. Show that the quotient A/B is abelian.

6. Let G be a group acting on a set A. Show that if $a, b \in A$ are such that $b = g \cdot a$ for some $g \in G$, then

$$\mathrm{Stab}(b) = g.\,\mathrm{Stab}(a).g^{-1} = \{ghg^{-1} : h \in \mathrm{Stab}(a)\}.$$

Deduce that if G acts transitively on A and $a \in A$ then the kernel of the action is $\bigcap_{g \in G} g\,\mathrm{Stab}(a)g^{-1}$.

7. Let

$$D_8 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$$

be the dihedral group with 8 elements, with $r^4 = s^2 = 1$, $sr = r^3 s$, and $sr^2 = r^2 s$.
Let $H = \langle s \rangle$ be the subgroup of D_8 generated by s. List the left cosets of H in D_8 and fix a labelling of these cosets.
Consider the action of D_8 by left multiplication on the set of left cosets $(D_8/H)_{\text{left}}$ of H in D_8 defined by the formula

$$\sigma \cdot (\tau H) = \sigma\tau H.$$

Exhibit the image of each element of D_8 under the permutation representation

$$\pi_H : D_8 \to S_4$$

of D_8 in S_4 obtained from the above action of D_8 on the set $(D_8/H)_{\text{left}}$ and the above fixed labelling of elements of $(D_8/H)_{\text{left}}$. Deduce that this representation is faithful and S_4 contains a subgroup isomorphic to D_8.